Personal Information

Name: **Anton Vinogradov**

Student ID: **19205240**

Email: **anton.vinogradov@ucdconnect.ie**

Submitted on: **22.04.2020**

Data Context

The data used in this research consists of three datasets collected from Dutch primary schools. The first dataset contains pupils' gender, gathered through questionnaires filled out by students and pupils (they could tick the to-be-grouped with). The second dataset contains additional student information, such as gender and the number of years they have been in the school. The third dataset contains teachers' information, such as teaching experience, education level, and belonging to a relative-student pairings and meaningful relational gender distribution.

These datasets originate from real-world school administration records. Teachers collect student preferences through questionnaires, and then, based on the collected information, they form classes. The goal is to create balanced classes, maximize satisfaction and attentiveness to the teachers' instructions. The data is stored in a Excel file and provided to researchers for analysis.

Data Description

The data used in this research is from a collection of three datasets collected from Dutch primary schools. The first dataset contains student gender, gathered through questionnaires filled out by students and pupils (they could tick the to-be-grouped with). The second dataset contains additional student information, such as gender and the number of years they have been in the school. The third dataset contains teachers' information, such as teaching experience, education level, and belonging to a relative-student pairings and meaningful relational gender distribution.

These datasets originate from real-world school administration records. Teachers collect student preferences through questionnaires, and then, based on the collected information, they form classes. The goal is to create balanced classes, maximize satisfaction and attentiveness to the teachers' instructions. The data is stored in an Excel file and provided to researchers for analysis.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import networkx as nx
import sys
from collections import Counter
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.tree.export_graphviz import export_graphviz
from IPyWidgets import widgets, interact, interact_manual
from IPyWidgets.widgets.widget_selection import TabWidget
from IPyWidgets.widgets.widget_selection.ipystatmap.ipystatmap import IpyStatmap
from IPyWidgets.widgets.widget_selection.ipystatmap.ipystatmap.widgets.widget_selection.ipystatmap.ipystatmap import IpyStatmapWidget

# Loading and cleaning data
df = pd.read_excel("../data/Preferences.xlsx", sep='|')
student_info = pd.read_excel("../data/StudentInfo.xlsx", sep='|')
teacher_info = pd.read_excel("../data/TeacherInfo.xlsx", sep='|')

# Renaming columns for convenience
df.rename(columns={'Student Number': 'Student', 'Preference 1': 'Preference_1', 'Preference 2': 'Preference_2', 'Preference 3': 'Preference_3', 'Preference 4': 'Preference_4', 'Preference 5': 'Preference_5'}, inplace=True)
student_info.rename(columns={'Student Number': 'Student', 'Gender': 'Gender', 'Number of years in school': 'Years'}, inplace=True)
teacher_info.rename(columns={'Teacher Number': 'Teacher', 'Gender': 'Gender', 'Teaching experience': 'Experience', 'Education level': 'Education', 'Belonging to a relative-student pairings': 'Relative', 'Meaningful relational gender distribution': 'Gender_Distribution'}, inplace=True)

# Dropping duplicates
student_info.drop_duplicates(inplace=True)
teacher_info.drop_duplicates(inplace=True)

# Checking data
print("Preferences shape:", df.shape)
print("Student info shape:", student_info.shape)
print("Teacher info shape:", teacher_info.shape)

# Displaying first few rows of each dataset
print("\nPreferences head:\n", df.head()
print("\nStudent info head:\n", student_info.head()
print("\nTeacher info head:\n", teacher_info.head()

# Checking for missing values
print("\nMissing values in preferences:\n", df.isnull().sum()
print("\nMissing values in student info:\n", student_info.isnull().sum()
print("\nMissing values in teacher info:\n", teacher_info.isnull().sum()

# Checking for duplicate entries
print("\nDuplicate preferences:", df.duplicated().sum()
print("\nDuplicate student info:", student_info.duplicated().sum()
print("\nDuplicate teacher info:", teacher_info.duplicated().sum()

# Preview of the data
print("\nPreferences sample:\n", df.sample(5)
print("\nStudent info sample:\n", student_info.sample(5)
print("\nTeacher info sample:\n", teacher_info.sample(5)

# Basic statistics
print("\nPreferences statistics:\n", df.describe()
print("\nStudent info statistics:\n", student_info.describe()
print("\nTeacher info statistics:\n", teacher_info.describe()

# Checking unique values in categorical columns
print("\nUnique student genders:", student_info['Gender'].unique()
print("\nUnique teacher genders:", teacher_info['Gender'].unique()
print("\nUnique teacher experience levels:", teacher_info['Experience'].unique()
print("\nUnique teacher education levels:", teacher_info['Education'].unique()
print("\nUnique relative relations:", teacher_info['Relative'].unique()
print("\nUnique gender distributions:", teacher_info['Gender_Distribution'].unique()

# Checking relationship between teacher experience and education
print("\nExperience vs Education:\n", pd.crosstab(teacher_info['Experience'], teacher_info['Education']).sample(5)

# Checking relationship between teacher gender and relative relations
print("\nGender vs Relative:\n", pd.crosstab(teacher_info['Gender'], teacher_info['Relative']).sample(5)

# Checking relationship between student gender and number of years in school
print("\nGender vs Years:\n", pd.crosstab(student_info['Gender'], student_info['Years']).sample(5)
```

```
Preferences shape: (1000, 6)
Student info shape: (1000, 3)
Teacher info shape: (100, 6)

Preferences head:
   Student  Group  Preference_1  Preference_2  Preference_3  Preference_4  Preference_5
0      1    G01            G02           G03           G04           G05           G06
1      2    G02            G01           G03           G04           G05           G06
2      3    G03            G01           G02           G04           G05           G06
3      4    G04            G01           G02           G03           G05           G06
4      5    G05            G01           G02           G03           G04           G06

Student info head:
   Student Gender  Years
0      1      F      7
1      2      M      7
2      3      M      7
3      4      M      7
4      5      F      7

Teacher info head:
   Teacher Gender  Experience  Education  Relative  Gender_Distribution
0      1      F           5        MBO         N                    F
1      2      M           5        MBO         N                    M
2      3      F           5        MBO         N                    F
3      4      M           5        MBO         N                    M
4      5      F           5        MBO         N                    F

Missing values in preferences:
 Student       0
Group         0
Preference_1  0
Preference_2  0
Preference_3  0
Preference_4  0
Preference_5  0
dtype: int64

Missing values in student info:
 Student  0
Gender   0
Years    0
dtype: int64

Missing values in teacher info:
 Teacher              0
Gender               0
Experience           0
Education            0
Relative             0
Gender_Distribution  0
dtype: int64

Duplicate preferences: 0

Duplicate student info: 0

Duplicate teacher info: 0

Preferences sample:
     Student  Group  Preference_1  Preference_2  Preference_3  Preference_4  Preference_5
     Student  Gender  Years
     Teacher  Gender  Experience  Education  Relative  Gender_Distribution
```